Temporal Biased Streaming Submodular Optimization

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Data Streams

- Big data:
  - online social networks
  - Internet of things
  - mobile devices
  - ...

- Data streams: “3V” challenges
  - Volume: in TBs even PBs
  - Velocity: K/s (≈9500 tweets/sec)
  - Variety: texts, images, video, numerical data, etc.

information overload

Data Stream Summarization

- **Goal**: use a carefully chosen subset of items to represent the stream at any time point
- **Challenges**:
  - data items arrives at a very fast speed
  - each data item can only be visited once
  - random access to the entire data is not allowed
  - only a small fraction of the data can be loaded into memory

surveillance camera

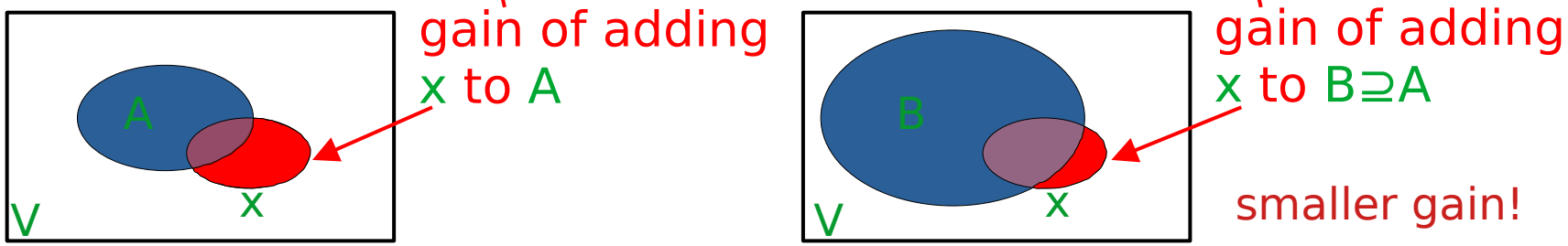
summary

Submodularity

- Submodularity is a natural model for
  - representativeness, informativeness, diversity, coverage
- A set function $f: 2^V \mapsto \mathbb{R}_{\geq 0}$ is submodular if

$$\underline{f(A \cup \{x\}) - f(A)} \geq \underline{f(B \cup \{x\}) - f(B)}$$

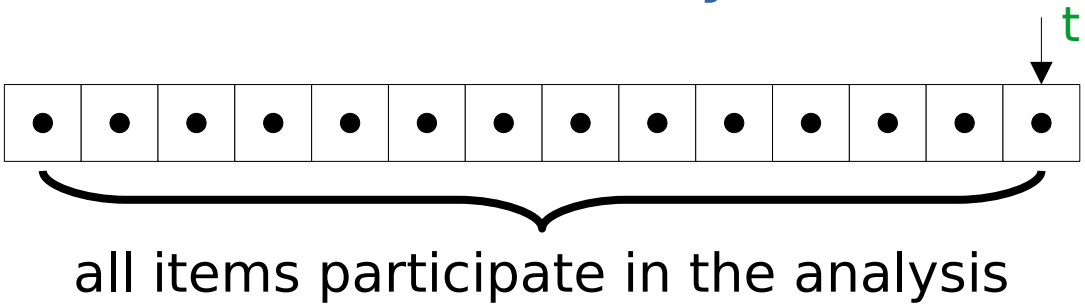
for all $A \subseteq B \subseteq V$ and $x \in V \setminus B$.



- Captures the diminishing returns property

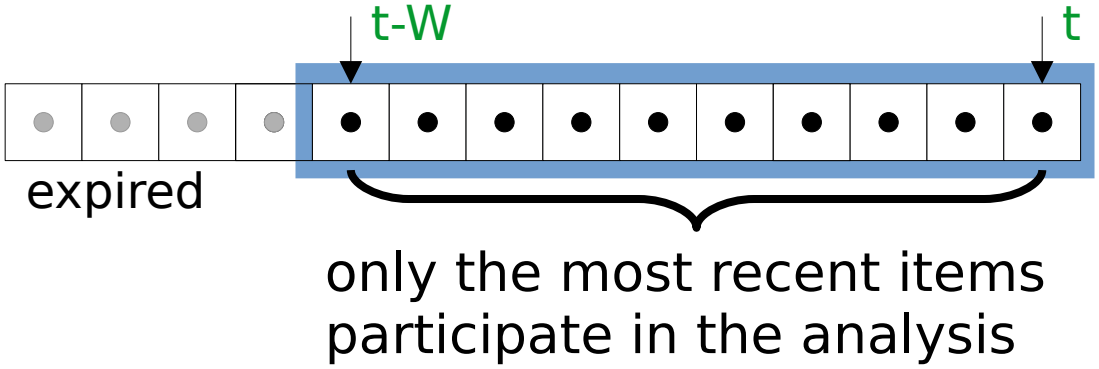
Streaming Submodular Optimization (SSO)

- SSO for insertion-only streams [KDD'14]

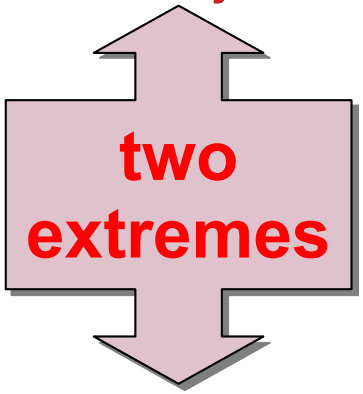


cons:
- all items are treated equally regardless of how outdated they are

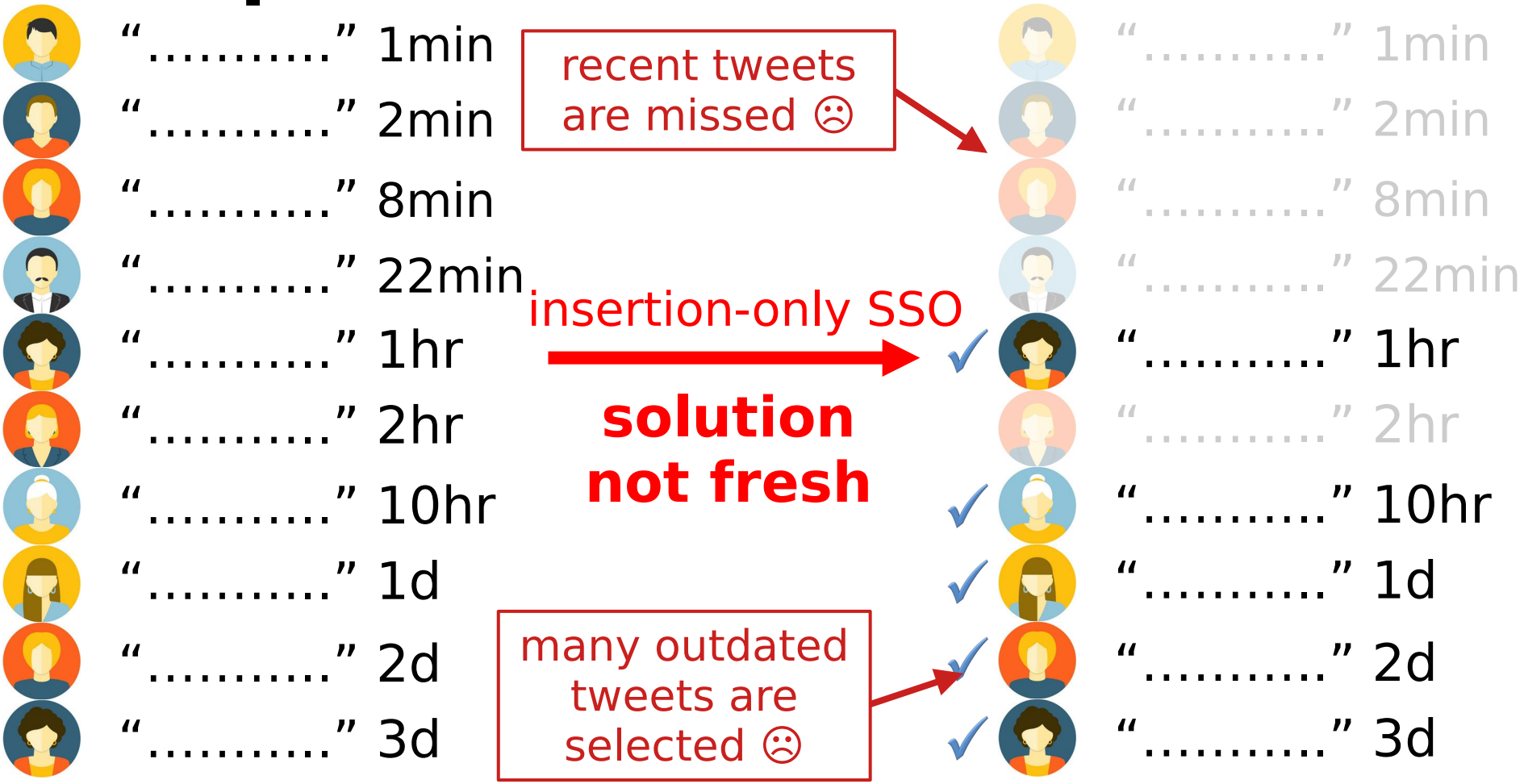
- SSO for sliding-window streams [WWW'17]



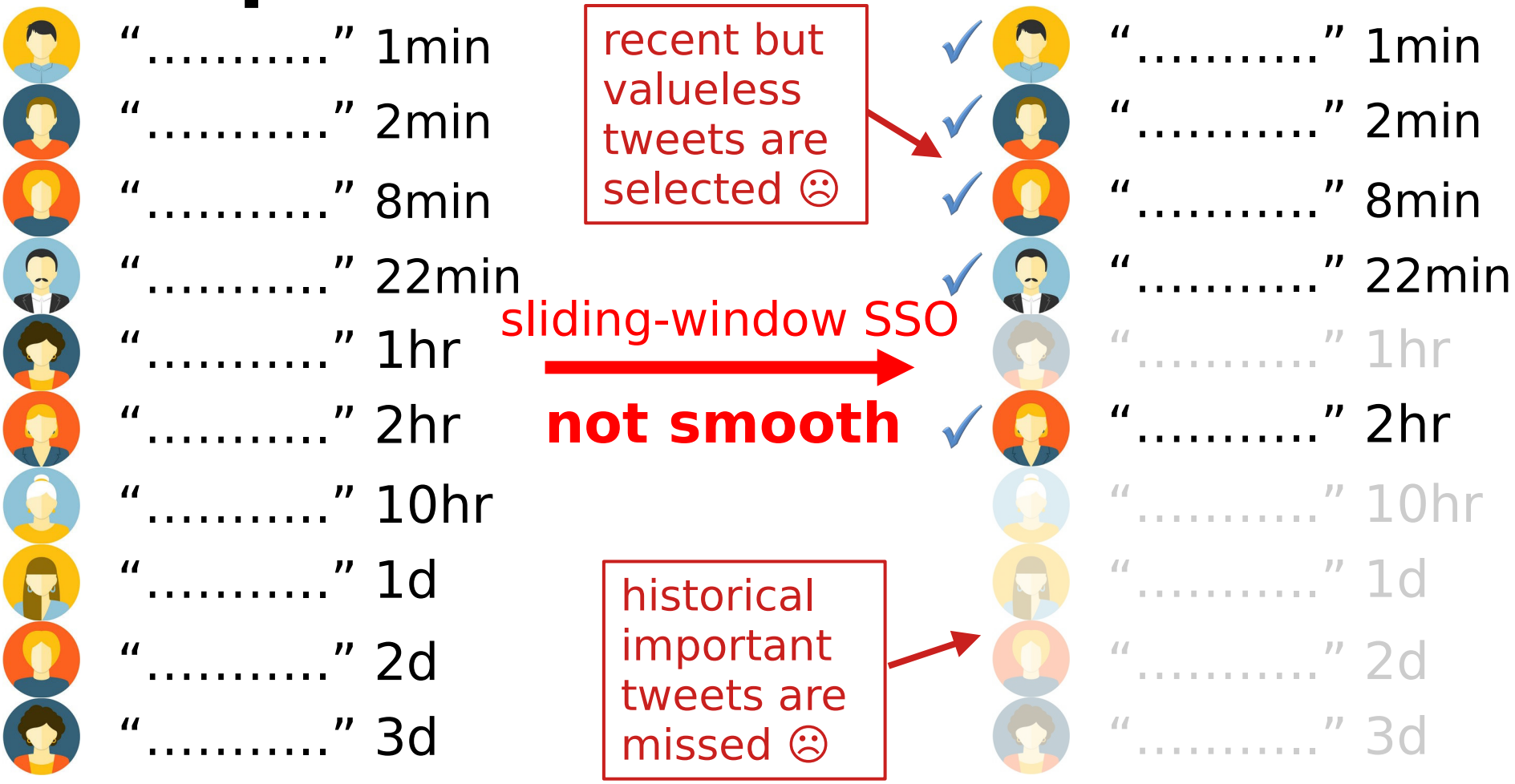
cons:
- abruptly forgets all past data which may be still important



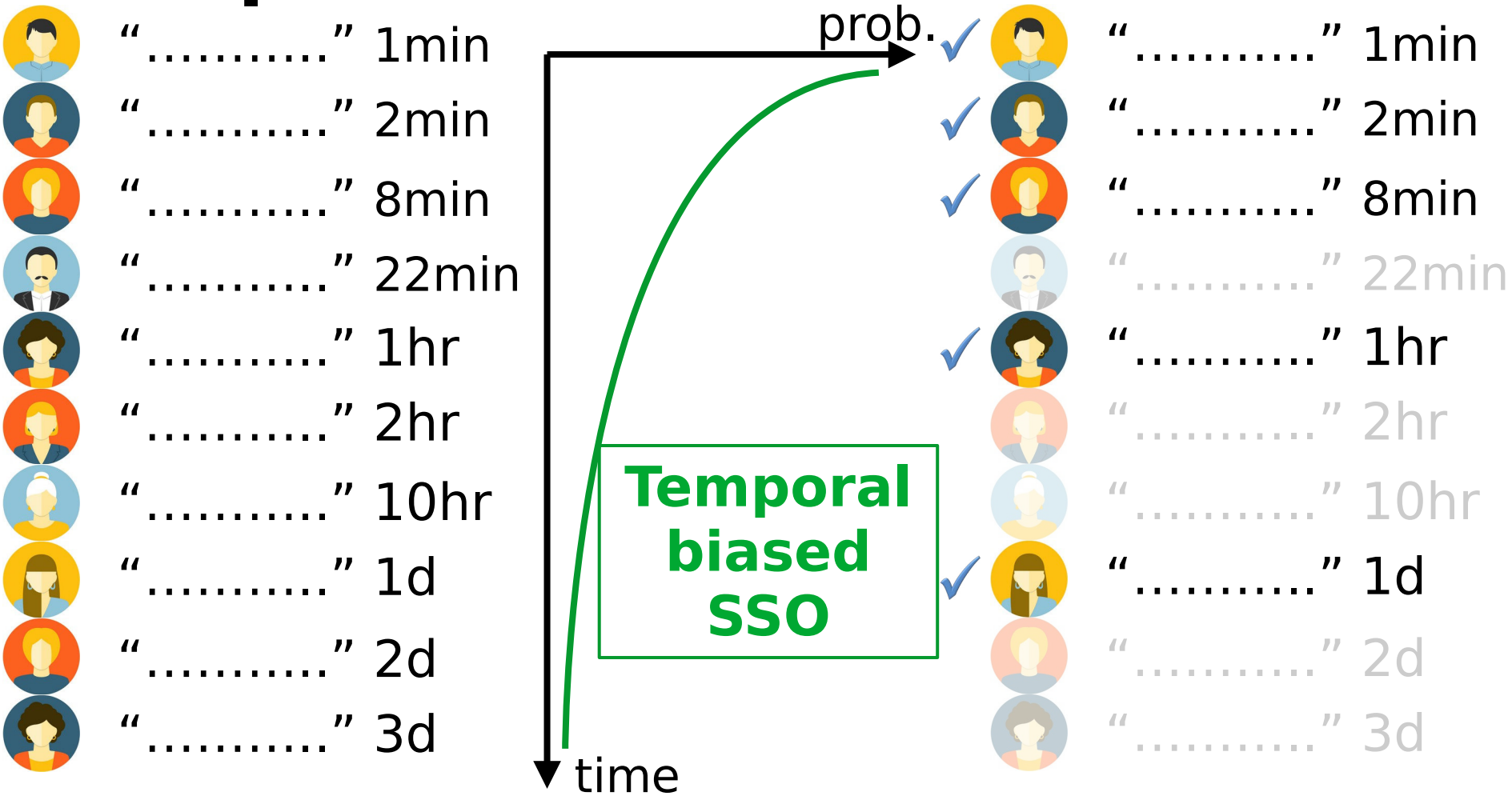
Example: Tweets Recommendation



Example: Tweets Recommendation



Example: Tweets Recommendation

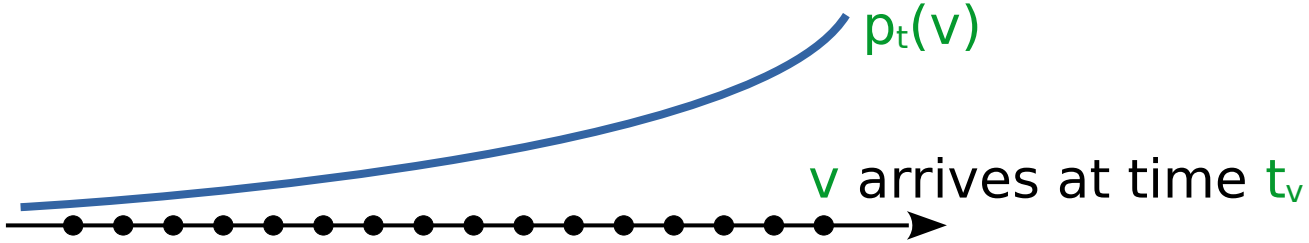


Outline

- Background & Motivation
- ☞ **TBSSO Problem Formulation**
- Algorithms
- Experiments
- Conclusion

Temporal Biased Stream Model

- Each item v participates in the analysis with a probability $p_t(v)$ decreasing over time.



- and $p_t(v) \triangleq h(t - t_v)$ where $h: Z \mapsto [0,1]$ assigns an item of age x a participation probability $h(x)$, called the decay function, e.g.,
  - $h(x) = p_0 e^{-\lambda x}$, i.e., an exponential decay function

Temporal Biased SSO Problem

TBSSO Problem formulation:

- **Given** a stream of items $\mathbb{S}_t = \{v \in V: t_v \leq t\}$ with decay function $h(x)$,
- **Want** to find k items $S \subseteq V$ that maximize $\mathbb{E}_h[f(S|\mathbb{S}_t)]$ at any query time t.

- The TBSSO problem generalizes previous settings:
  - If $h(x) = 1$, $\forall x$, then becomes insertion-only SSO.
  - If $h(x) = 1$ for $x \leq W$, and $h(x) = 0$ otherwise, then becomes sliding-window SSO.

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How to Calculate $\mathbb{E}_h[f(S|\mathbb{S}_t)]$?

Example:

Suppose $S=\{a,b\}$, then

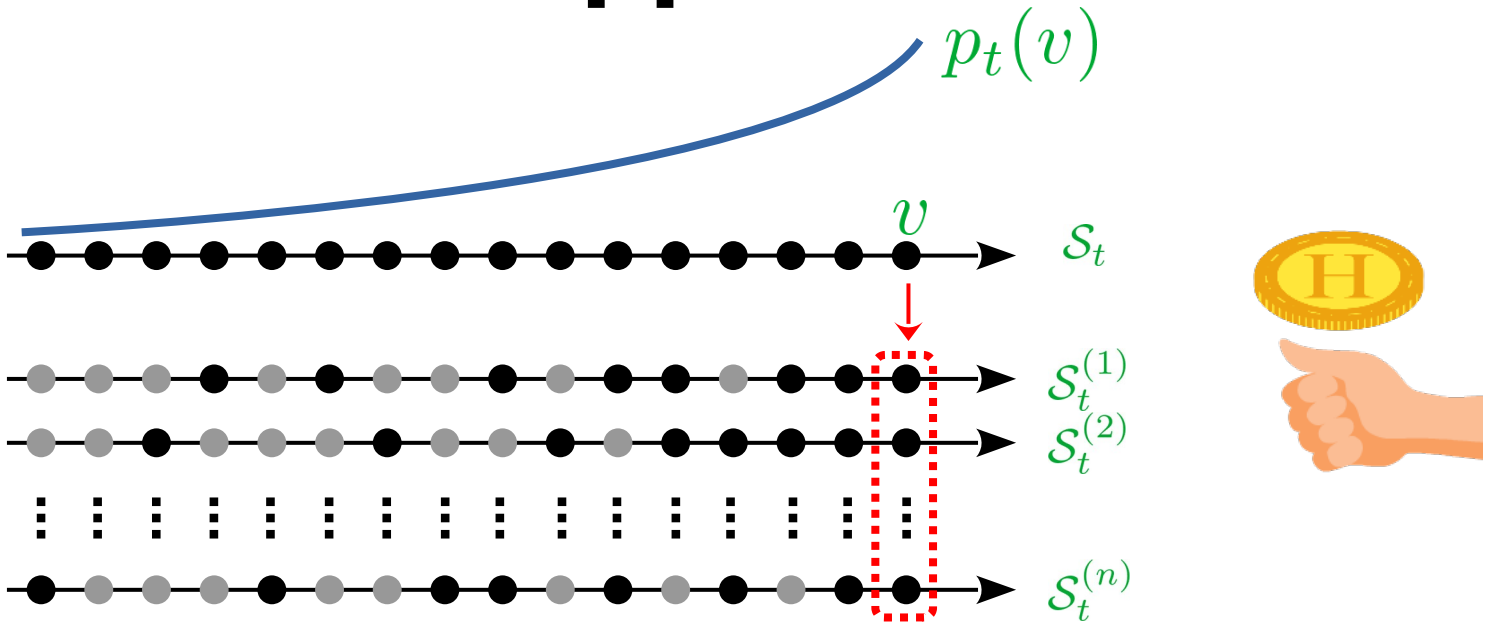
$$\mathbb{E}_h[f(\{a,b\}|\mathbb{S}_t)] = \boxed{p_t(a)p_t(b)f(\{a,b\})}$$ both a and b participate in the analysis

$$+ \boxed{p_t(a)(1 - p_t(b))f(\{a\})}$$ only a participates in the analysis

$$+ \boxed{(1 - p_t(a))p_t(b)f(\{b\})}$$ only b participates in the analysis

- Exactly calculating $\mathbb{E}_h[f(S|\mathbb{S}_t)]$ needs $O(2^{|S|})$ oracle calls,
  - one oracle call refers to one evaluation of f.
- **Too expensive!**

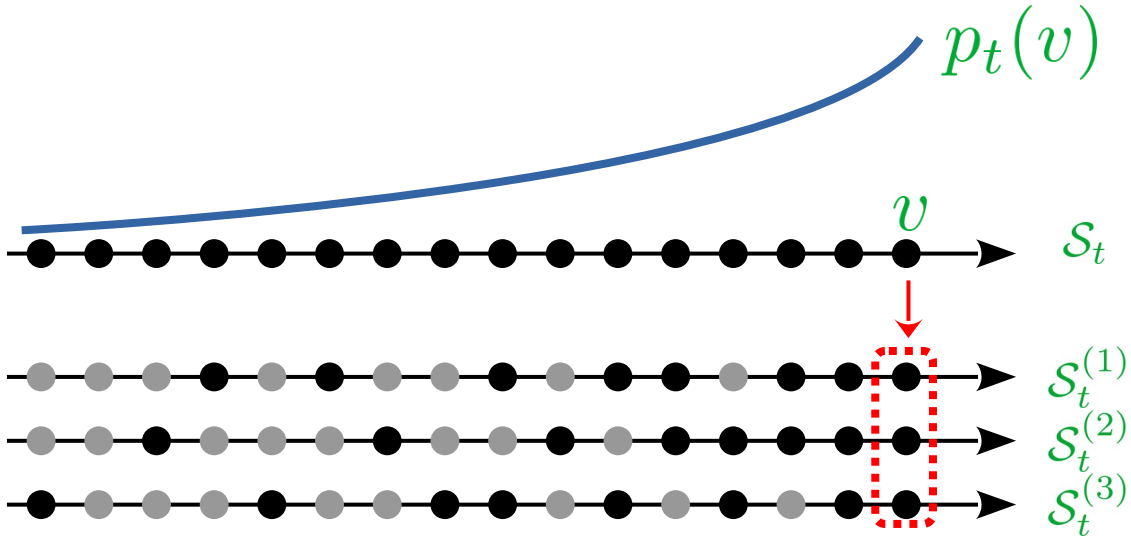
Monte-Carlo Approximation



$$F(S) \triangleq \frac{1}{n}\sum_{i=1}^{n} f(S \cap \mathcal{S}_t^{(i)}) \xrightarrow{a.s.} \mathbb{E}_h[f(S|\mathcal{S}_t)] \text{ as } n \to \infty$$

require n oracle calls, n << $2^{|S|}$

Bernoulli Set (B-Set)

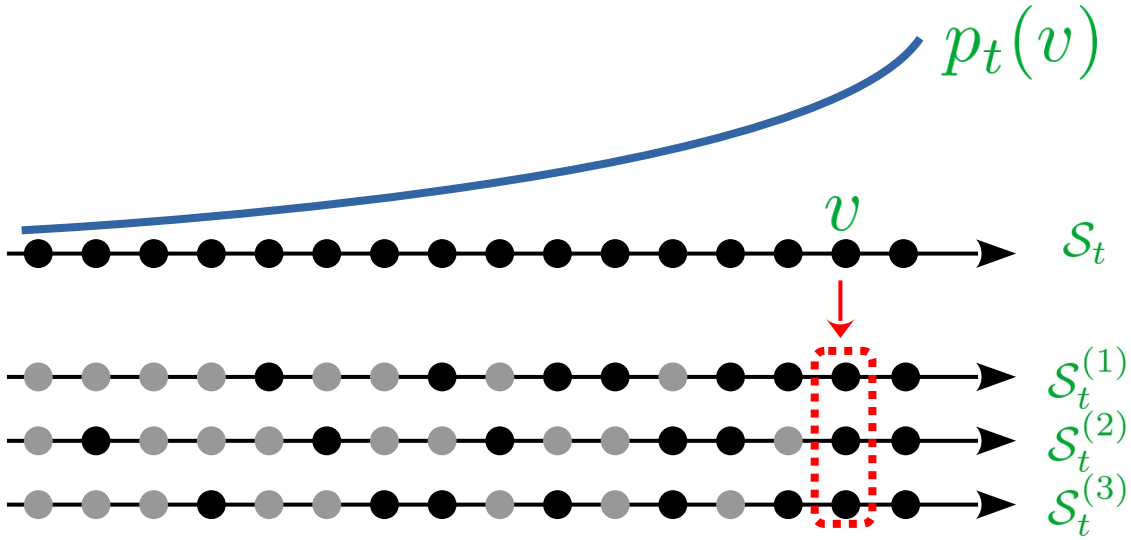


$$I_0(v) = \{1, 2, 3\}$$

- B-set of an item v at time $t_v + l$:

$$I_l(v) \triangleq \{i \colon v \in \mathcal{S}_t^{(i)} \wedge t_v + l = t\}$$

Bernoulli Set (B-Set)



$$I_1(v) = \{1, 2, 3\}$$

- B-set of an item v at time $t_v + l$:

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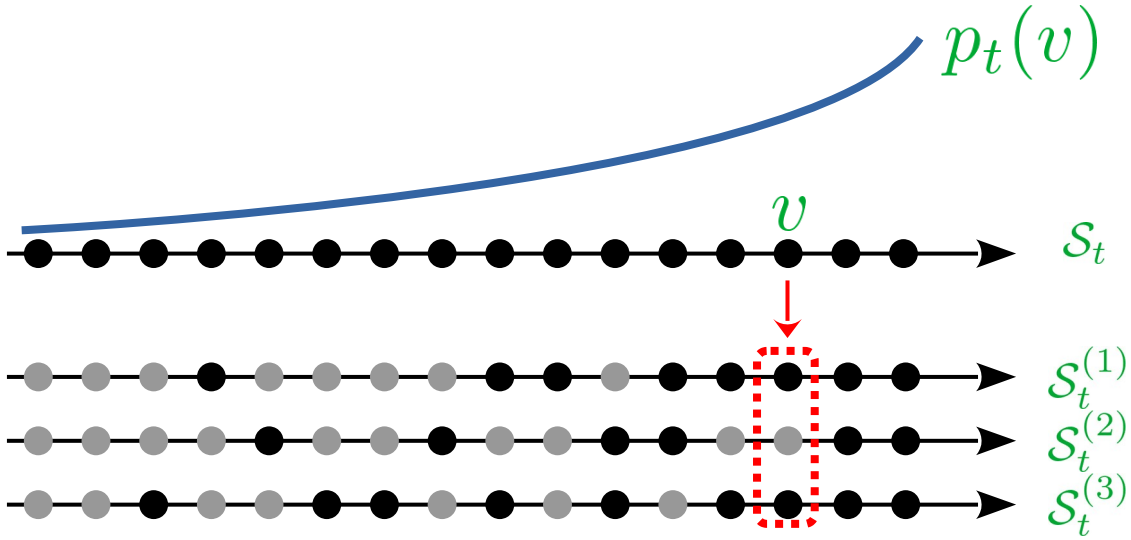
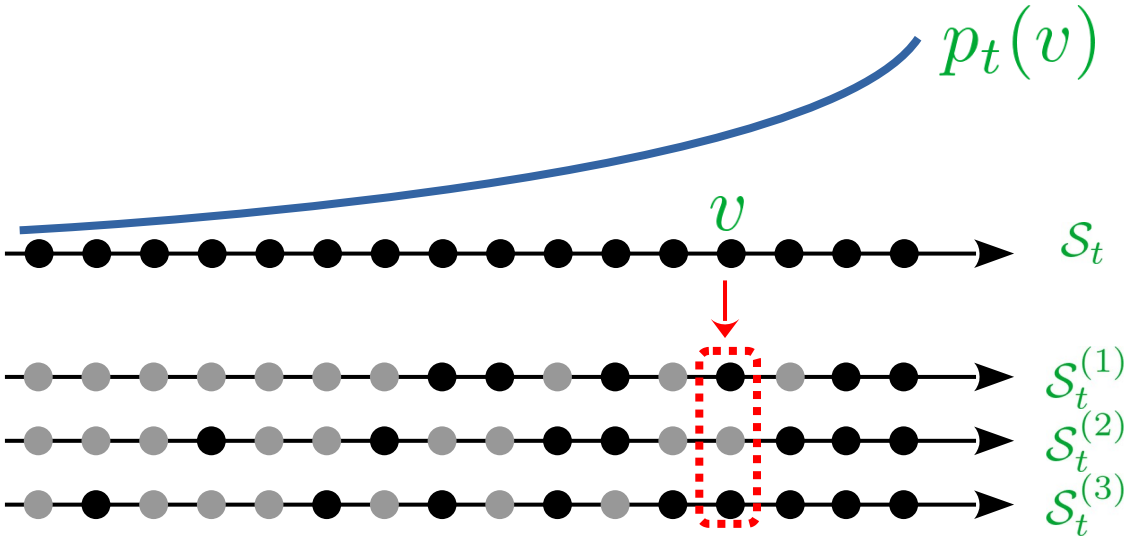


$$I_2(v) = \{1, 3\}$$

- B-set of an item v at time $t_v + l$:

$$I_l(v) \triangleq \{i \colon v \in \mathcal{S}_t^{(i)} \wedge t_v + l = t\}$$

Bernoulli Set (B-Set)



$$I_3(v) = \{1, 3\}$$

- B-set of an item v at time $t_v + l$:

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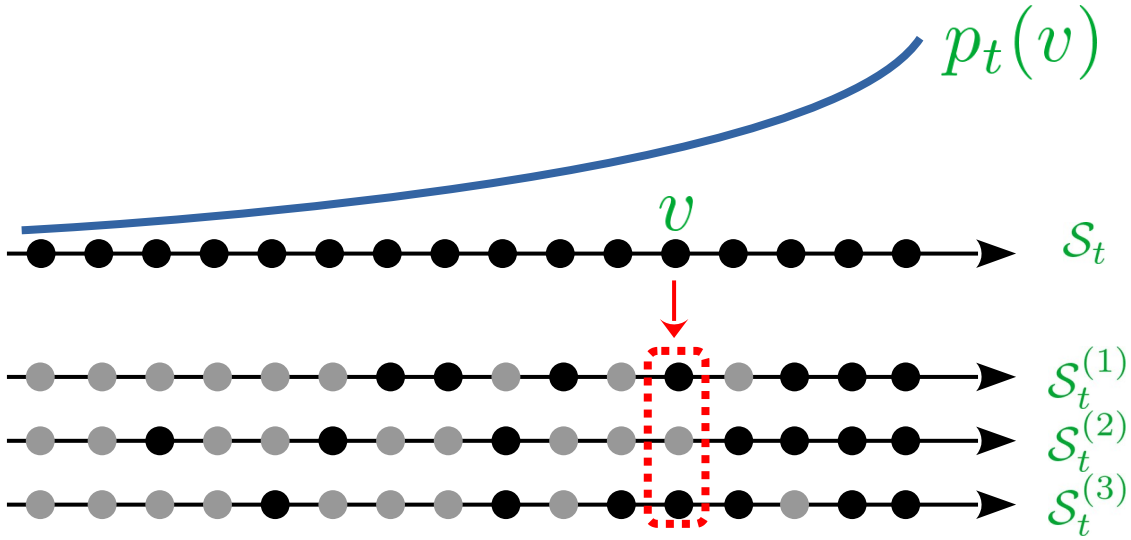
Bernoulli Set (B-Set)

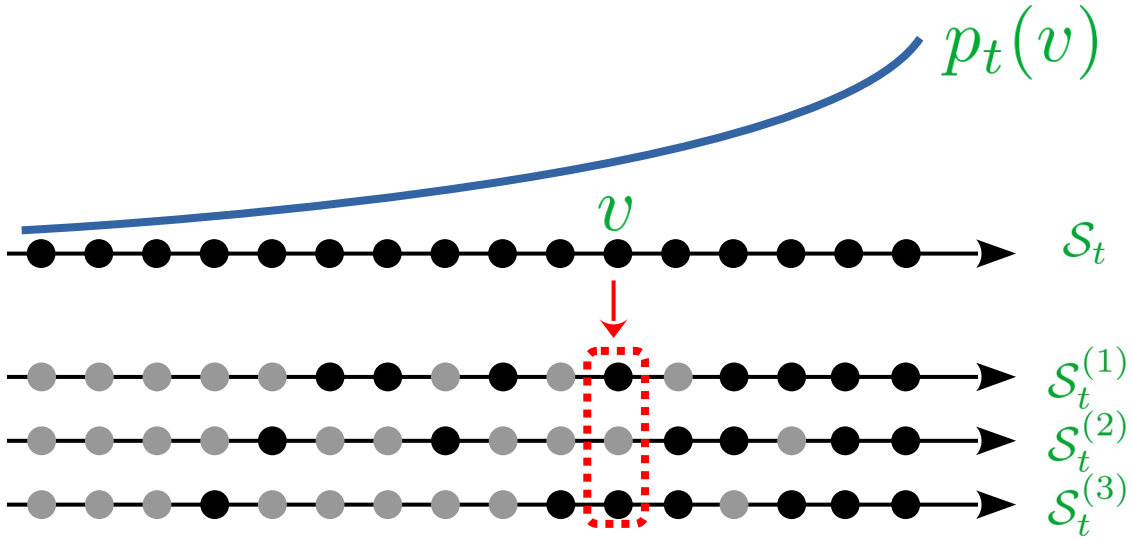


$$I_4(v) = \{1, 3\}$$

- B-set of an item v at time $t_v + l$:

$$I_l(v) \triangleq \{i \colon v \in \mathcal{S}_t^{(i)} \wedge t_v + l = t\}$$

Bernoulli Set (B-Set)

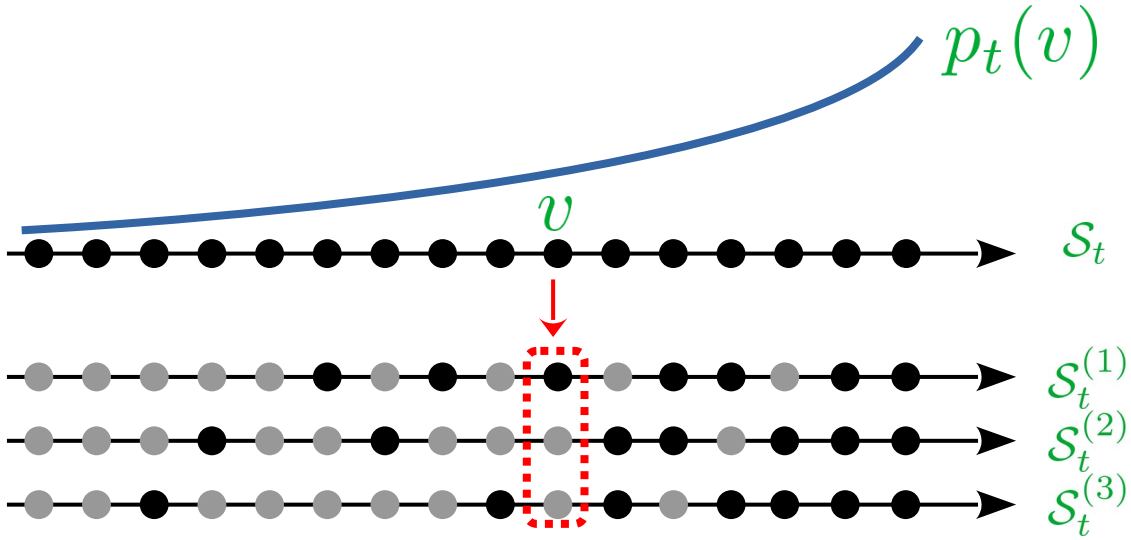


$$I_5(v) = \{1, 3\}$$

- B-set of an item v at time $t_v + l$:

$$I_l(v) \triangleq \{i \colon v \in \mathcal{S}_t^{(i)} \wedge t_v + l = t\}$$

Bernoulli Set (B-Set)

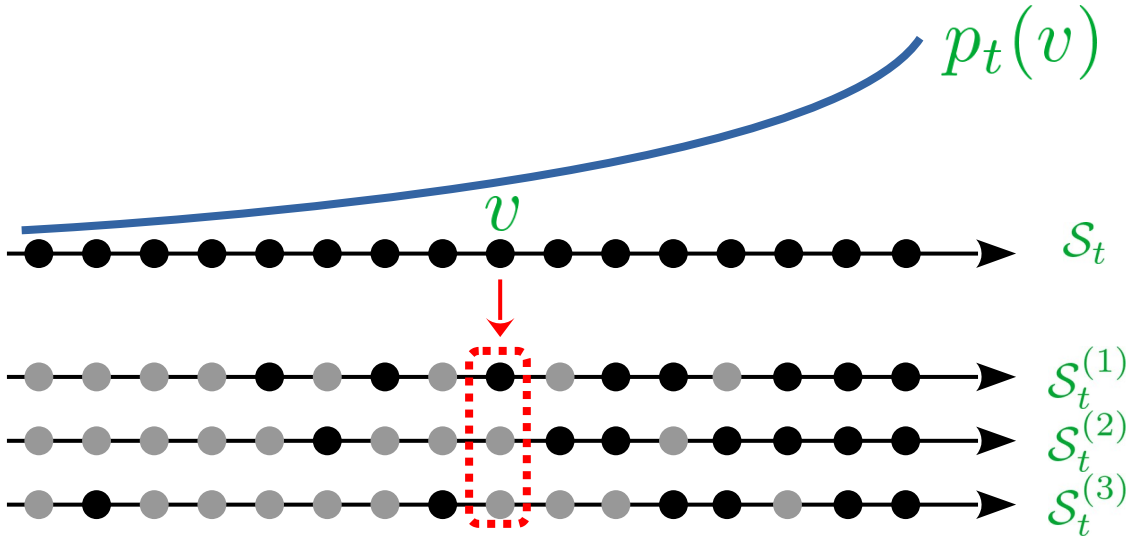


$$I_6(v) = \{1\}$$

- B-set of an item v at time $t_v + l$:

$$I_l(v) \triangleq \{i \colon v \in \mathcal{S}_t^{(i)} \wedge t_v + l = t\}$$

Bernoulli Set (B-Set)

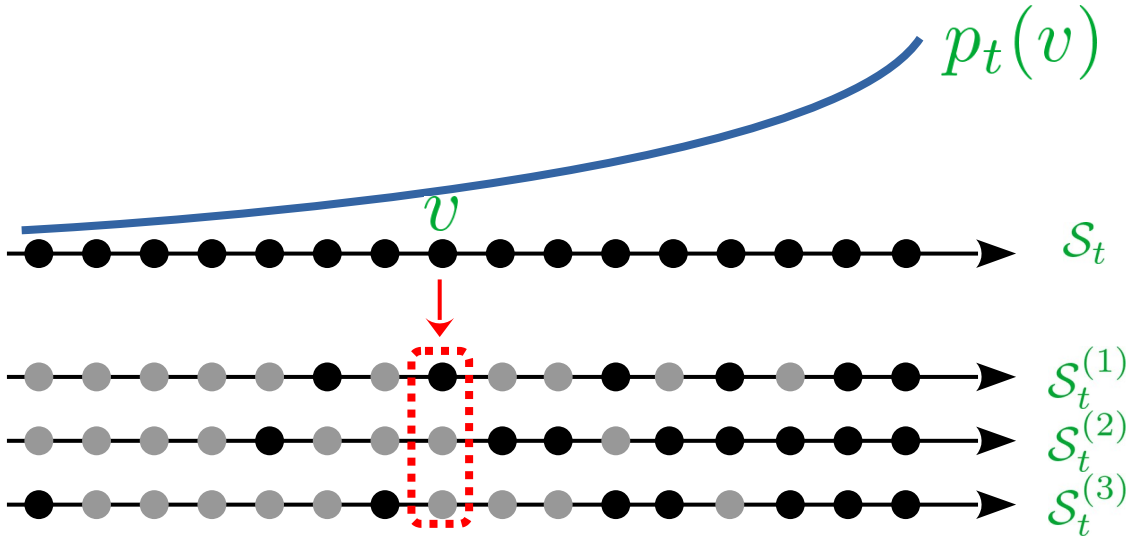


$$I_7(v) = \{1\}$$

- B-set of an item v at time $t_v + l$:

$$I_l(v) \triangleq \{i \colon v \in \mathcal{S}_t^{(i)} \wedge t_v + l = t\}$$

Bernoulli Set (B-Set)

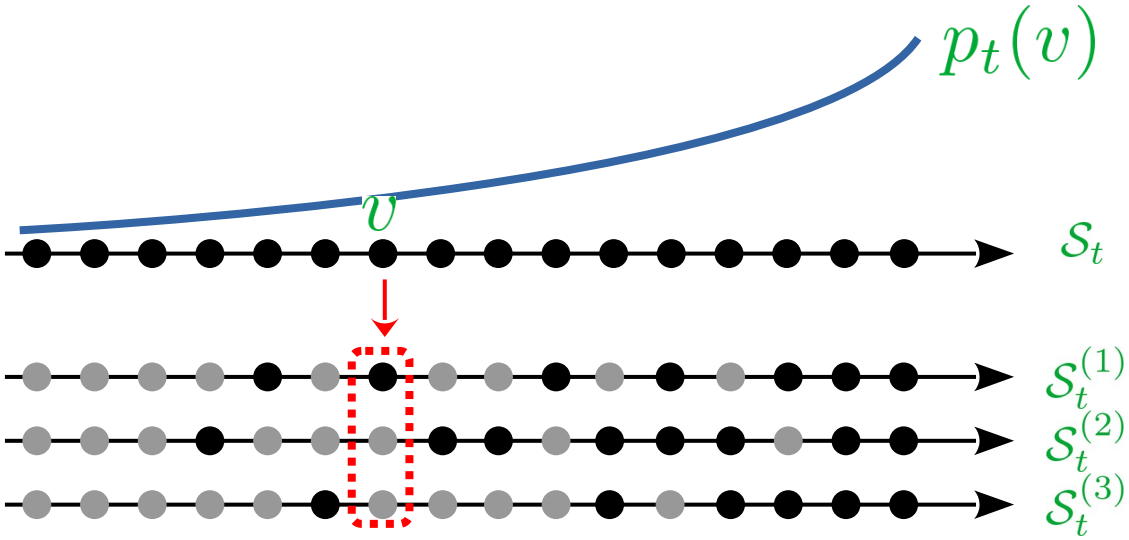


$$I_8(v) = \{1\}$$

- B-set of an item v at time $t_v + l$:

$$I_l(v) \triangleq \{i \colon v \in \mathcal{S}_t^{(i)} \wedge t_v + l = t\}$$

Bernoulli Set (B-Set)



$$I_9(v) = \{1\}$$

- B-set of an item v at time $t_v + l$:

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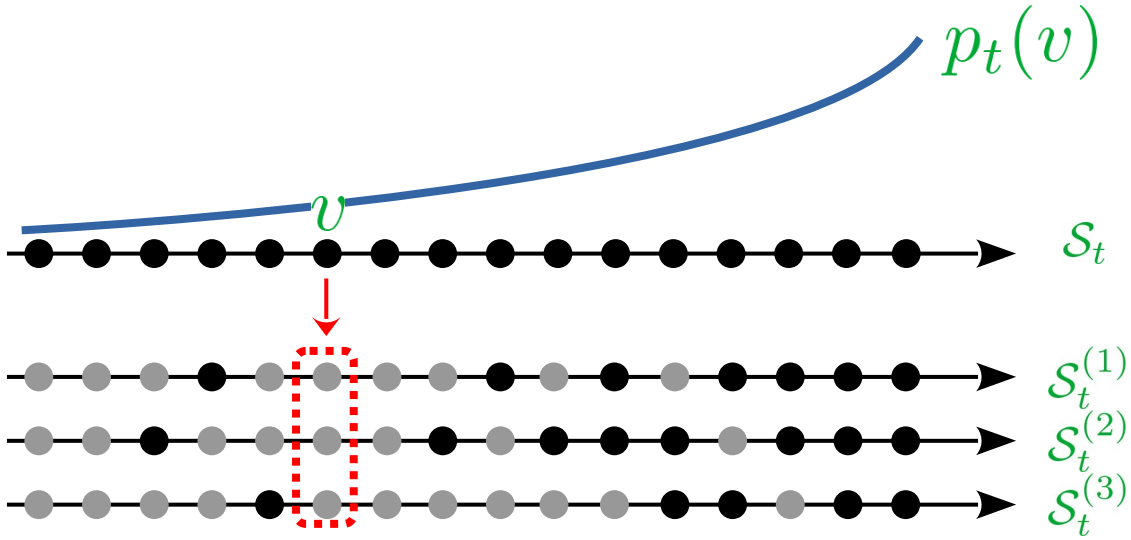
Bernoulli Set (B-Set)



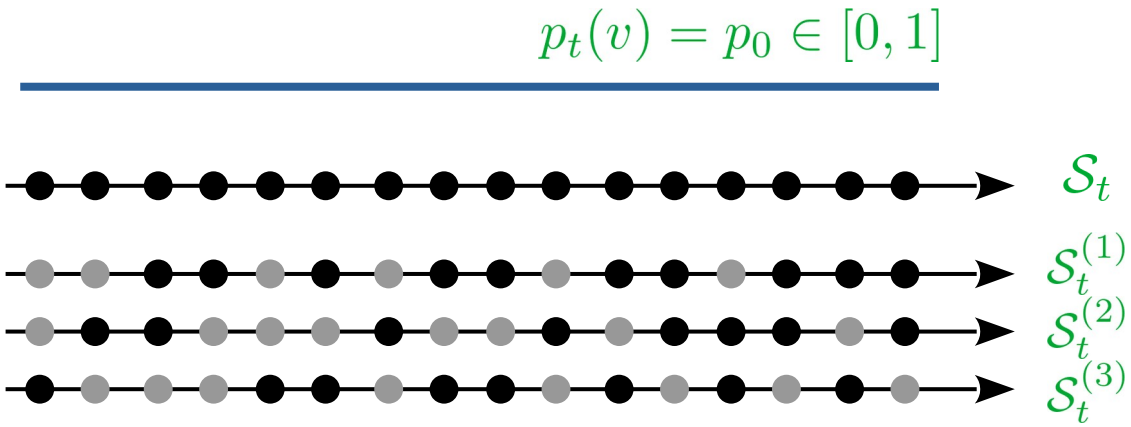
$I_{10}(v) = \emptyset$

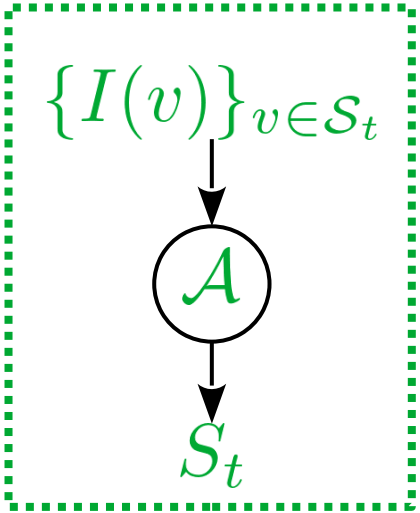
- B-set of an item v at time $t_v + l$:

$$I_l(v) \triangleq \{i \colon v \in \mathcal{S}_t^{(i)} \wedge t_v + l = t\}$$

- Eventually, a B-set will shrink to an empty set.
- B-set is another way to represent probabilistic decays.

The Non-Decaying Case

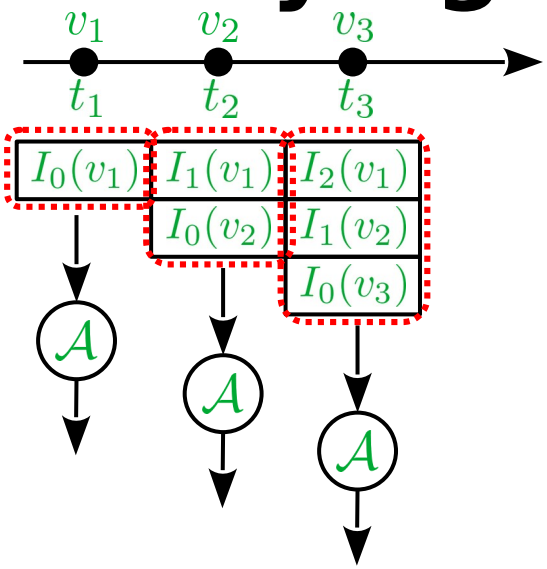




Atom

- Each item's B-set is a constant set.
- The stream becomes an insertion-only stream, where each item in the stream is a B-set, i.e., $\{I(v) : v \in \mathcal{S}_t\}$
- Many insertion-only SSO algorithms can be applied.
  - denote one implementation by Atom.

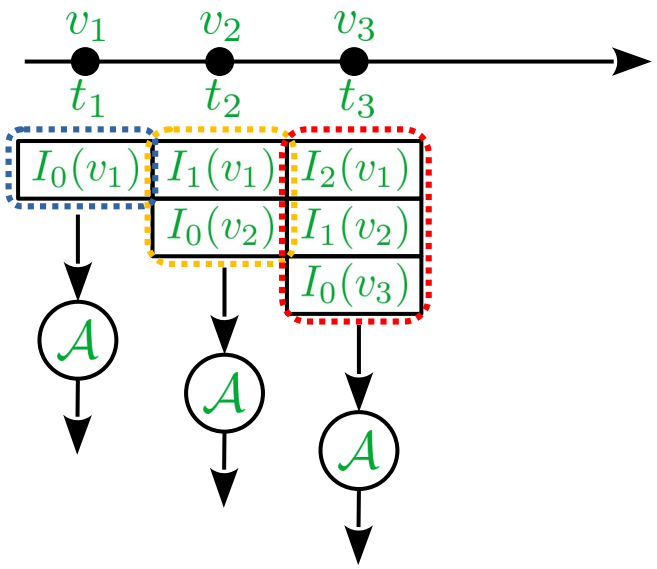
The Decaying Case

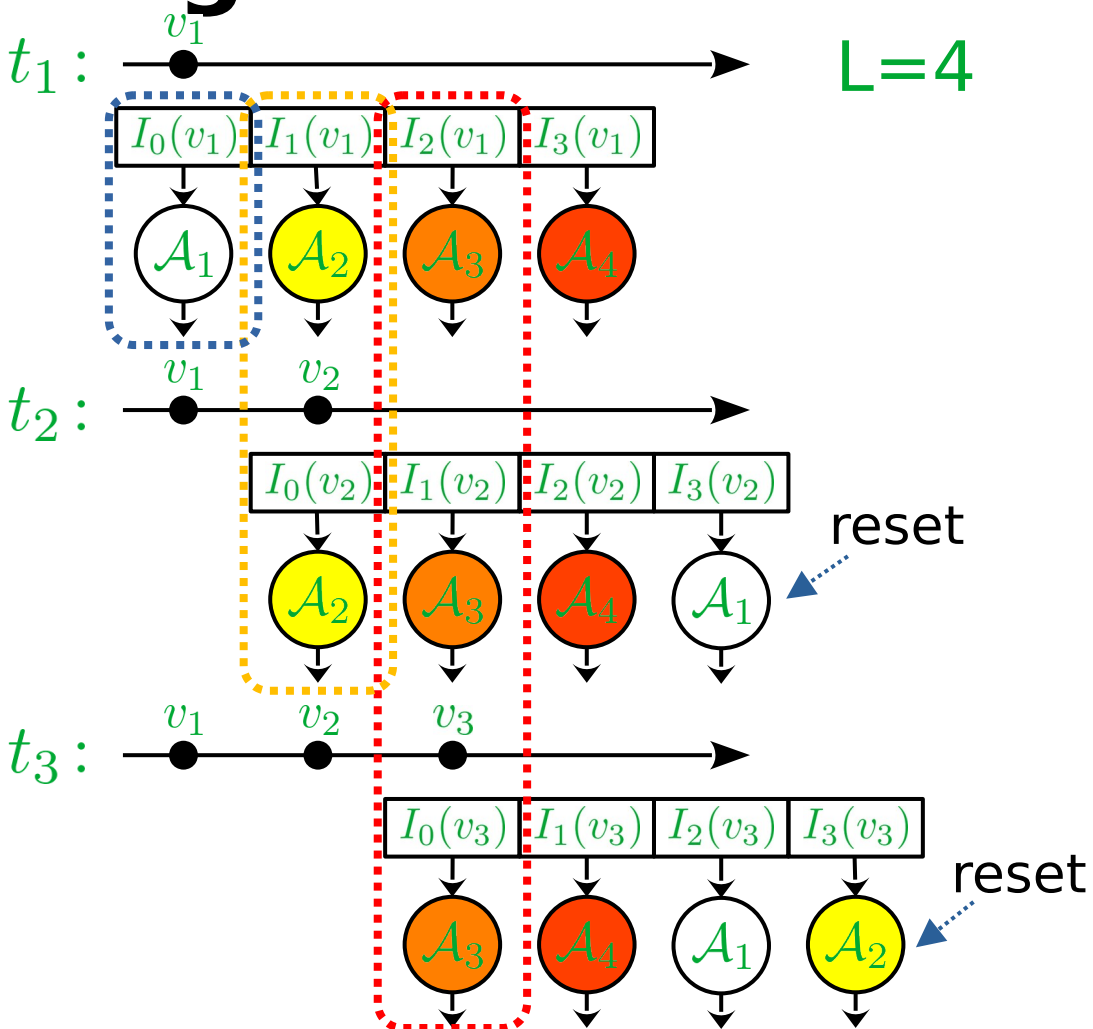


$$I_0(v_1) \supseteq I_1(v_1) \supseteq I_2(v_1) \supseteq \dots$$
$$I_0(v_2) \supseteq I_1(v_2) \supseteq \dots$$

- **Idea**: If we can feed the B-sets at each time point to an Atom instance, then the instance's output will be the solution of the TBSSO problem at the time point.
- **Challenge**: Each B-set is shrinking, how to process these evolving B-sets in a streaming fashion?

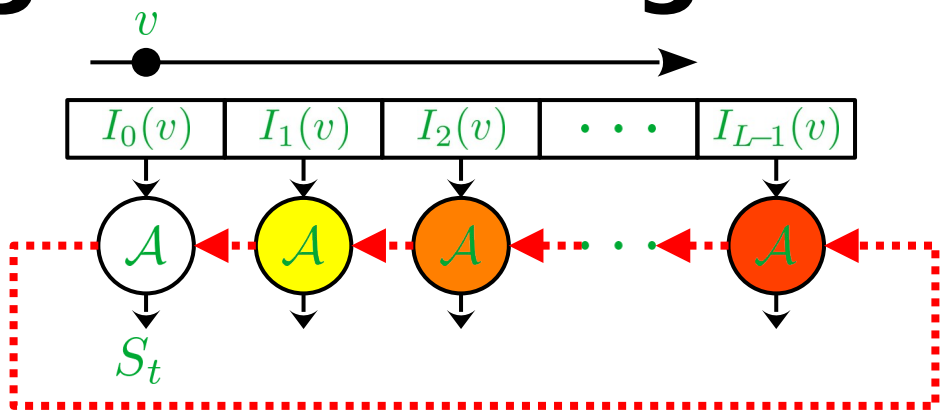
A Basic Algorithm





- **Assume** each item has at most L non-empty B-sets.

Basic Alg: Processing and Shifting

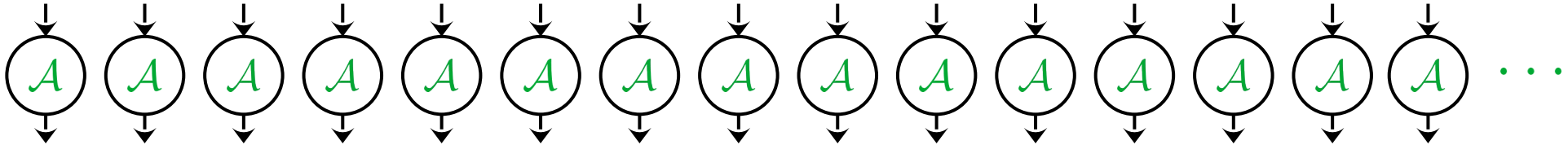


reset the first instance and then shift

- For each item v, obtain its B-sets $\{\mathrm{I}_l(\mathrm{v}): 0 \leq l < \mathrm{L}\}$;
- Feed these B-sets to L Atom instances, respectively;
- Reset the first instance and circularly shift these instances before processing the next item;
- The first Atom instance's out is always the solution.

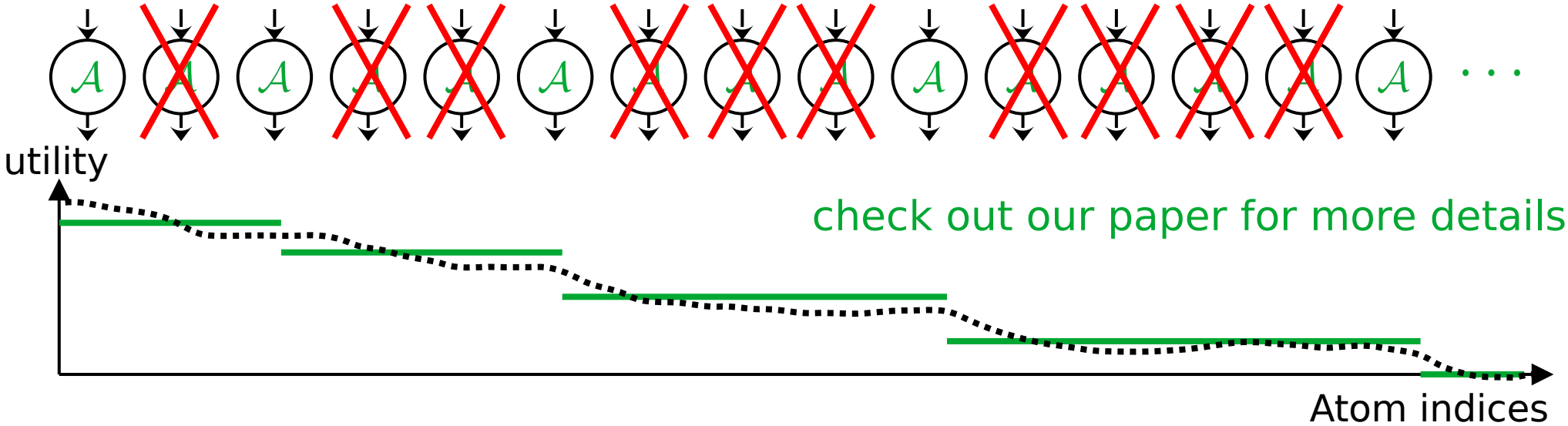
Limitations of the Basic Algorithm

- What if L is very large, e.g., $L = \infty$?
- **Weakness**: Maintaining a large number of Atom instances will incur too much RAM and CPU overloads.



A Faster Algorithm

- **Idea**: remove redundant Atom instances
  - trick 1: group same B-sets into one segment, each segment needs only one Atom instance;
  - trick 2: remove Atom instances that have close outputs
- Similar to use a histogram to approximate a curve.



Summary

Algorithm	Approx. Ratio	Update Time	Space
Atom	α	β	γ
Basic Alg.	α	$O(L\beta)$	$O(L\gamma)$
Fast Alg.	$\alpha(1-\varepsilon)/2$	$O(\beta\varepsilon^{-1}\log k)$	$O(\gamma\varepsilon^{-1}\log k)$

- For example, if Atom is implemented by adapting SieveStreaming [KDD'14], then
  - $\alpha=1/2 - \varepsilon$
  - $\beta=O(n\varepsilon^{-1}\log k)$
  - $\gamma=O(nk\varepsilon^{-1}\log k)$

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- ☞ **Experiments**
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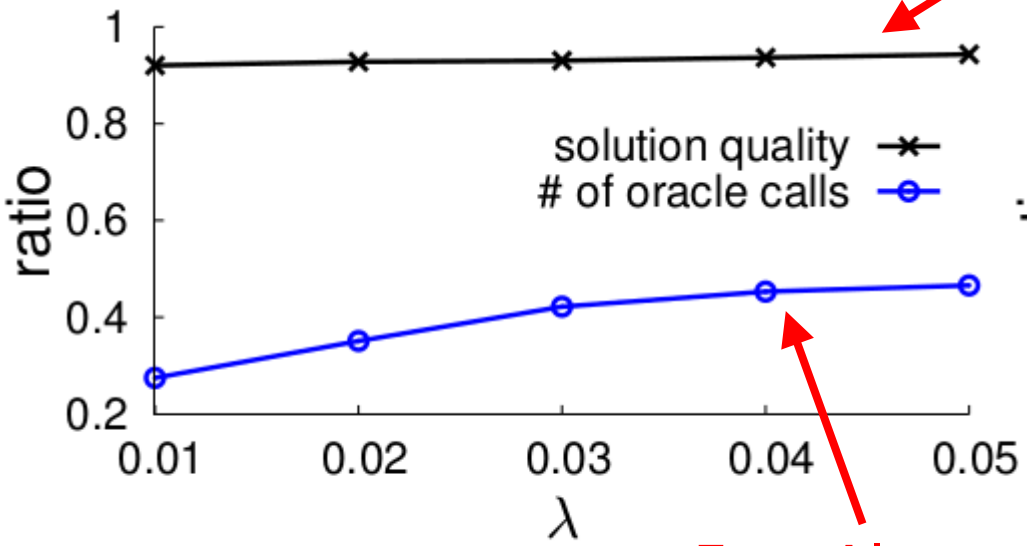
Dataset

data stream	element	length	time period
DBLP [9]	author	372K	1936 - 2018
MemeTracker [21]	article	714K	1/1/2009 - 31/1/2009
StackOverflow [26]	question	2.9M	1/1/2015 - 1/3/2016
US2020-Tweets [11]	tweet	1.7M	15/10/2020 - 8/11/2020

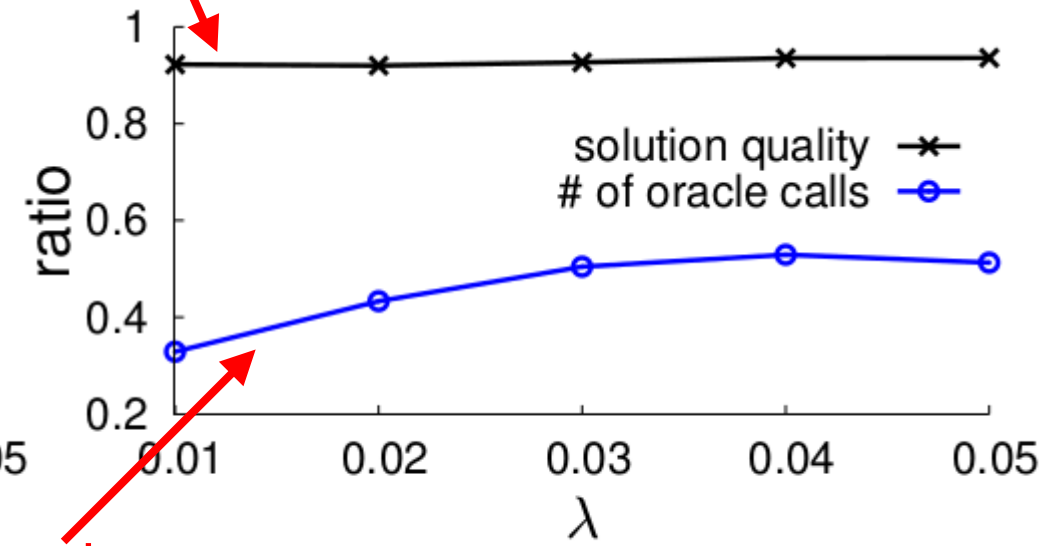
- **Goal**: maintain k items that have the maximum coverage, i.e., $f(S) = |\cup_{v \in S} v|$ where each item is a set.
- **Decay function**: $h(x) = e^{-\lambda x}$, $\lambda \geq 0$.
- Baselines:
  - Greedy: uses the lazy evaluation trick, non-streaming;
  - Random: randomly select k items.

Basic Alg. vs Fast Alg.

similar solution quality



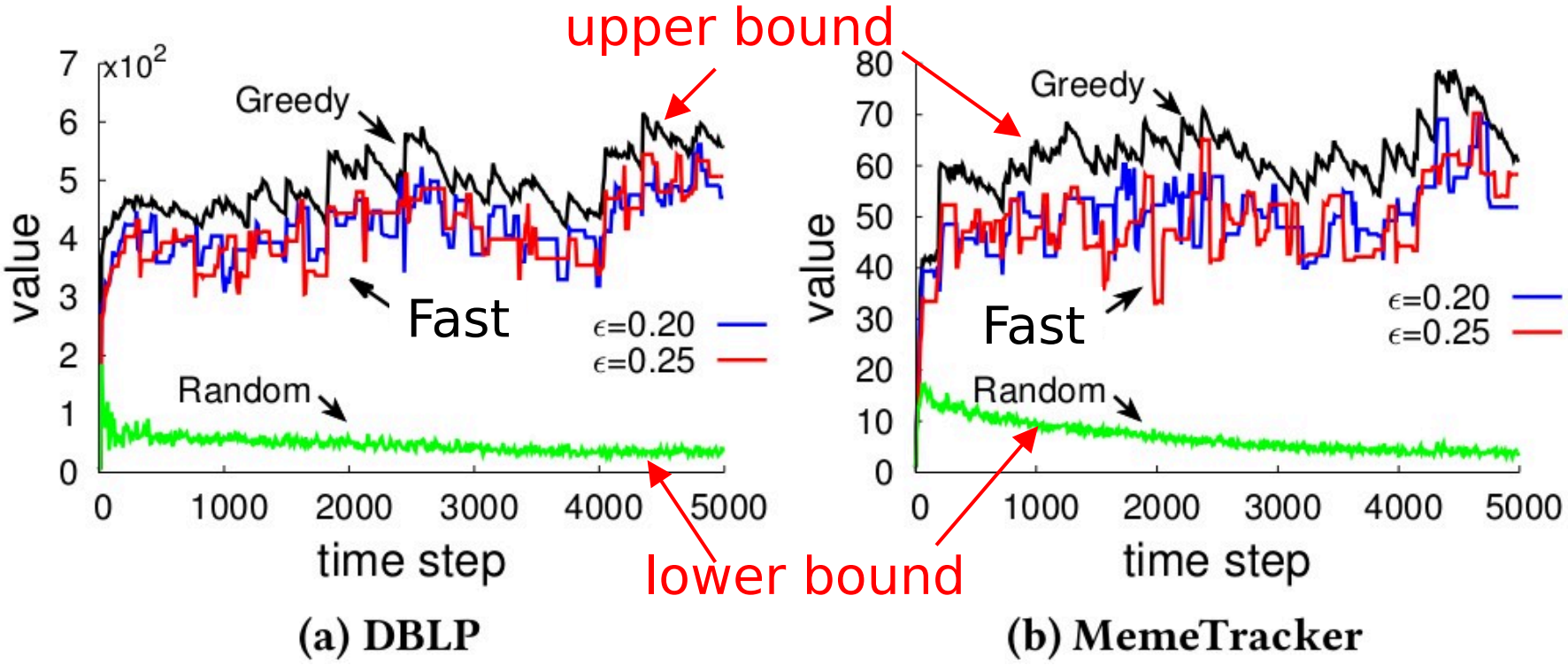
(a) DBLP



(b) MemeTracker

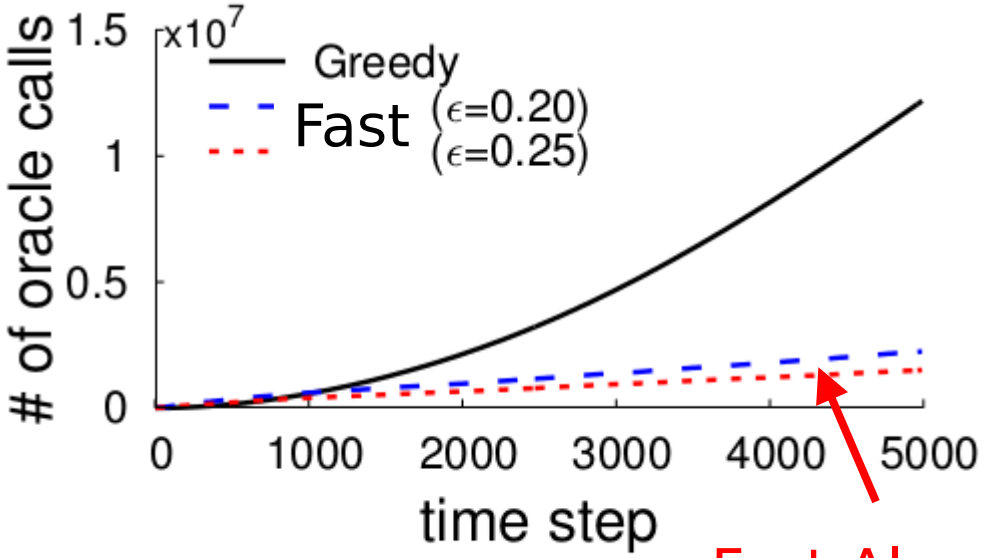
Fast Alg needs fewer oracle calls than Basic Alg

Solution Quality



- The fast algorithm is close to the solution quality of Greedy, and is much better than the Random.

Computational Efficiency



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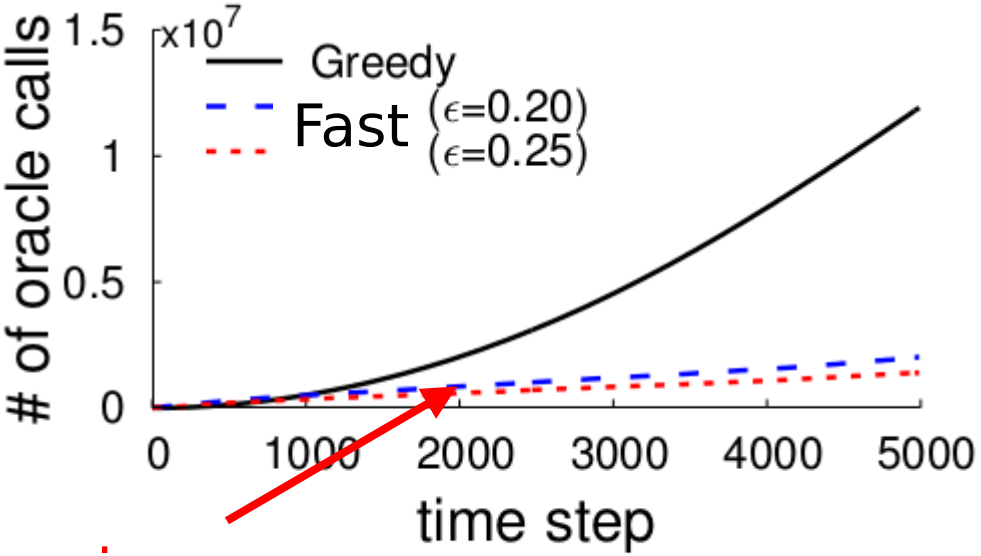

(b) MemeTracker

Fast Alg. requires much fewer oracle calls than Greedy

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Conclusion

- A novel Temporal Biased Streaming Submodular Optimization problem.
- Can be reduced to SSO over insertion-only streams.
- The proposed algorithm can find near-optimal solutions with much fewer costs than baselines.

Algorithm	Approx. Ratio	Update Time	Space
Atom	α	β	γ
Basic Alg.	α	$O(L\beta)$	$O(L\gamma)$
Fast Alg.	$\alpha(1-\varepsilon)/2$	$O(\beta\varepsilon^{-1}\log k)$	$O(\gamma\varepsilon^{-1}\log k)$

Thanks for Listening!